Elementary graph algorithms: exercises

22-E01 Representations and zero weights

An undirected simple graph has 6 vertices and 9 edges stored as ordinary unsorted adjacency lists. Find the total adjacency entries, the time bound for complete BFS, and matrix space. For a directed weighted graph, explain why A[u,v]=0 cannot unambiguously mean both an absent edge and a legitimate zero-weight edge. Give an unambiguous representation.

22-E02 BFS on six vertices

Use directed lists a:[b,c],b:[c],c:[a],d:[b,e],e:[d],f:[e]. Run BFS from d. List every removed vertex, every end-of-iteration queue, all distances, and all predecessors. Identify a directed edge whose finite endpoint distances differ by more than one and explain why this respects the BFS theorem.

22-E03 SCCs and crossing transposed components

The only directed edges are a→b,b→a,b→c,c→d. Run the first DFS in a,b,c,d order with alphabetized adjacency lists. Give tin/tout, the transposed lists, the correct second-pass candidate order, and the SCCs. Which vertices would an arbitrary search from d in G^T reach? Why does that not justify omitting the root order?

22-E04 DFS edge types on the same graph

Use the graph from 22-E02 with the stated outer and adjacency orders. List every tin/tout and π, and classify all 8 edges. Explain why a→c is neither tree nor cross. Can tin/tout alone distinguish tree from forward edges?

22-E05 Valid topological orders and remaining vertices

For a→c,b→c,c→d, test orders a,b,c,d; b,a,c,d; and a,c,b,d. Show the FIFO Kahn queue starting with a,b. Then add d→b and d→z, with no edge out of z. Give Kahn's output prefix, the remaining vertices, and one cycle. Does every remaining vertex lie on a cycle?

22-E06 White paths versus ordinary reachability

For a:[b,c],b:[],c:[b], run DFS in a,b,c order. List discovery and finish events, and the white vertices immediately before c is discovered. Although c→b exists, why is b not a descendant of c? Identify the essential white-path theorem condition.

Elementary graph algorithms: exercises (continued)

22-E07 Shortest-path counts and bit cost

A directed unit-weight simple graph has edges $0\to1$,$0\to2$,$1\to3$,$2\to3$,$1\to4$,$2\to4$,$3\to5$,$4\to5$. Run BFS from 0 and list d and ways, including each addition into ways[5]. Why must counting include edges into an already discovered v? With $\Theta(n)$ layers, two vertices per layer, and all possible edges to the next layer, how do the path count and its bit length grow? State the cost model for a linear-time counting claim.

> Complete the derivation before consulting the separate solutions PDF.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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